



Denis Meleshkin

Energy Engineer

CONTACT

✉ melesh.denis@gmail.com

in [LinkedIn profile](#)

🌐 Open to relocation

🔗 [Portfolio](#)

TECHNICAL SKILLS

Energy systems design
2D/3D modelling
Heat-transfer calculations
Thermal & flow analysis
Entropy & exergy analysis
Life Cycle Assessment

SOFT SKILLS

Continuous learning
Adaptability
Collaboration
Problem-solving
Resource allocation
Academic research
Technical communication

SOFTWARE & TOOLS

MATLAB
Simulink / Simscape
ANSYS Fluent / CFX
OpenFOAM
Aspen HYSYS
AutoCAD / Inventor
Solid Edge / Works
CFturbo
Julia / Python
LaTeX / Git

LANGUAGES

English — fluent
Italian — conversational
Russian — native

INTERESTS

Travelling & exploring
Music & instruments
Hiking, surfing & skiing

Energy Engineer | Design & Analysis [🔗 My portfolio](#)

Driven to learn, adapt, and create innovative engineering solutions for a better, more sustainable future.

PROFILE

My background combines studying in different countries with experience across technical and business environments. A year of travelling and volunteering has strengthened my adaptability, independence, and ability to navigate dynamic environments. I aim to bring this perspective to my work while continuing to learn, adapt, and contribute to challenging projects.

EDUCATION

M.Sc. Energy Engineering

2023–2025

University of Padua, Padua, Italy

Thesis – Numerically investigated heat transfer enhancement using twisted tape turbulators in single-phase water flow, analysing the effects of geometric parameters and Reynolds number. Validated the numerical results against established heat-transfer correlations.

B.Sc. Refrigeration, Cryogenic Equipment and Life Support Systems

Bauman Moscow State Technical University

2019–2023

Thesis – Designed a helium cryogenic cooling system with a capacity of 2 kW at 100–150 K, performing the required thermodynamic calculations and equipment selection.

Elected as head of the study group and held this honorary position until graduation.

Engaged in research activities, resulting in publication and conference presentation [\[1\]](#) [\[2\]](#).

Diploma with honour.

WORK EXPERIENCE

Design Engineer

2022–2023

PJSC "Cryogenmash", Moscow, Russia

Under the supervision of a senior engineer, performed process simulations in Aspen HYSYS, equipment calculations and rating, and prepared technical documentation for an air separation unit for a metallurgical plant. Calculated the required system discharge pressure, supporting compressor selection while meeting technical requirements and minimising equipment costs.

Procurement technical engineer

2021–2022

LLC STATMED, Ivanovo, Russia

Managed international procurement of medical adhesive products, coordinating with suppliers, pricing and documentation. Led the introduction of a new adhesive tape product by negotiating its price, developing packaging design, and preparing product documentation.

INTERNSHIPS

Production practice

Amur Gas Processing Plant, Svobodny, Russia

During an internship at a large-scale gas processing plant, studied the natural gas separation line and investigated leakage issues in existing connections. Selected and experimentally tested four gasket types as replacement, identifying the best-performing option, which was subsequently installed.

Technological practice

NPO Nauka, Moscow, Russia

As part of an engineering team at an aircraft life-support systems manufacturer, developed the manufacturing process for a pressure regulator control mechanism and defined the complete process route.